# TidyTuesday Week 42

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Monday 12th October, 2026

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# 1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to the only recap on the internet brave enough to ask the hard-hitting question: what happens when you shove two completely unrelated economic series into a regression and press go? This week the random number gods served up a truly inspired pairing – the United States Balance on Services (a quarterly series so beloved it has since been DISCONTINUED, and so popular it scored a dazzling popularity rating of 1 out of whatever the maximum is) and, naturally, the Exchange Rate for Belize. Because if there's one thing that keeps American service exporters up at night, it's the Belizean dollar. Obviously.

And folks, we have a plot twist. Normally this is the part where I gleefully announce that the raw regression found a "significant" relationship that evaporates the instant you take away its security blanket of shared time-trend drift. The raw version did show up with a p-value of about 0.045, squeaking under the sacred 0.05 line like a student emailing an assignment at 11:59pm, explaining a whopping 9 percent of the variation – cute, but not exactly a relationship you'd introduce to your parents. The surprise is what happened after detrending: instead of collapsing in a heap, the relationship actually got stronger. The p-value dropped to about 0.018, the R-squared ticked up to 12 percent, and – here's the kicker – the coefficient flipped sign, going from positive in the raw run to negative once the trends were stripped out.

So in the rare-and-interesting column, we have a correlation that not only survived the detrending gauntlet but came out swinging harder than it went in. Dramatically, this is the closest thing to "evidence of a genuine link" this project ever produces, and I am contractually obligated to be mildly impressed. Does it mean anything? Did the balance on US services secretly dance to the tune of the Belizean exchange rate for half a century? Of course not. A sign-flipping, trend-dependent relationship between a dead services series and a small nation's currency is roughly as meaningful as finding your horoscope matches your grocery receipt. Nine to twelve percent of explained variance is also, let me gently remind everyone, mostly unexplained variance wearing a trench coat.

As always, the fine print: these series are paired at random, there is no theory, no hypothesis, no causal story, and absolutely nothing you should repeat at a dinner party as though it were fact. Spurious correlation is the house specialty here – it's the norm, not the exception – and the whole point of the detrending step is to catch the fakes. This week's result happens to be one of the sturdier ones we've stumbled into, which means precisely that it is slightly less laughable than usual. Treasure it. Next week we'll probably correlate cheese production with sunspot activity and feel just as confident.

## 2 Date: 2026-10-12

**Series ID: BOPBSVN**
This series is titled Balance on Services (DISCONTINUED) and has a frequency of Quarterly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1960-01-01 and the observation end date is 2014-01-01.The popularity of this series is 1.

**Series ID: XRNCUSBZA618NRUG**
This series is titled Exchange Rate (market+estimated) for Belize and has a frequency of Annual. The units are National Currency Units per US Dollar and the seasonal adjustment is Not Seasonally Adjusted.The observation start date is 1970-01-01 and the observation end date is 2023-01-01.The popularity of this series is 1.

### 2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

| | | | |
|---|---|---|---|
| **Dep. Variable:** | value_fred_XRNCUSBZA618NRUG | **R-squared:** | 0.090 |
| **Model:** | OLS | **Adj. R-squared:** | 0.069 |
| **Method:** | Least Squares | **F-statistic:** | 4.246 |
| **Date:** | Mon, 12 Oct 2026 | **Prob (F-statistic):** | 0.0454 |
| **Time:** | 20:28:58 | **Log-Likelihood:** | 34.026 |
| **No. Observations:** | 45 | **AIC:** | -64.05 |
| **Df Residuals:** | 43 | **BIC:** | -60.44 |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **const** | 1.9274 | 0.024 | 79.937 | 0.000 | 1.879 | 1.976 |
| **value_fred_BOPBSVN** | 0.0023 | 0.001 | 2.060 | 0.045 | 4.83e-05 | 0.004 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 17.591 | **Durbin-Watson:** | 0.425 |
| **Prob(Omnibus):** | 0.000 | **Jarque-Bera (JB):** | 23.370 |
| **Skew:** | -1.275 | **Prob(JB):** | 8.42e-06 |
| **Kurtosis:** | 5.441 | **Cond. No.** | 30.5 |

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

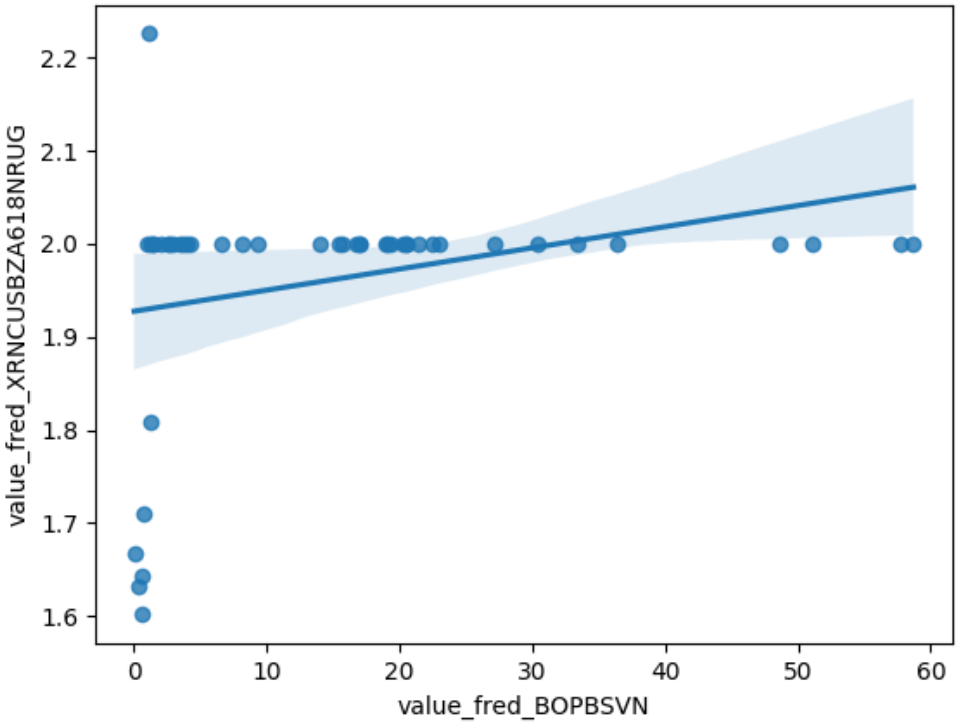



**Figure 1:** Raw Regression Plot for 2026-10-12

## 2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

| | | | |
|---|---|---|---|
| **Dep. Variable:** | value_fred_XRNCUSBZA618NRUG_detrended | **R-squared:** | 0.123 |
| **Model:** | OLS | **Adj. R-squared:** | 0.103 |
| **Method:** | Least Squares | **F-statistic:** | 6.041 |
| **Date:** | Mon, 12 Oct 2026 | **Prob (F-statistic):** | 0.0181 |
| **Time:** | 20:28:58 | **Log-Likelihood:** | 41.263 |
| **No. Observations:** | 45 | **AIC:** | -78.53 |
| **Df Residuals:** | 43 | **BIC:** | -74.91 |
| **Df Model:** | 1 | | |
| **Covariance Type:** | nonrobust | | |

| | **coef** | **std err** | **t** | **P> \|t\|** | **[0.025** | **0.975]** |
|---|---|---|---|---|---|---|
| **const** | 5.671e-15 | 0.015 | 3.84e-13 | 1.000 | -0.030 | 0.030 |
| **value_fred_BOPBSVN_detrended** | -0.0050 | 0.002 | -2.458 | 0.018 | -0.009 | -0.001 |

| | | | |
|---|---|---|---|
| **Omnibus:** | 8.915 | **Durbin-Watson:** | 0.588 |
| **Prob(Omnibus):** | 0.012 | **Jarque-Bera (JB):** | 16.952 |
| **Skew:** | 0.296 | **Prob(JB):** | 0.000208 |
| **Kurtosis:** | 5.948 | **Cond. No.** | 7.28 |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

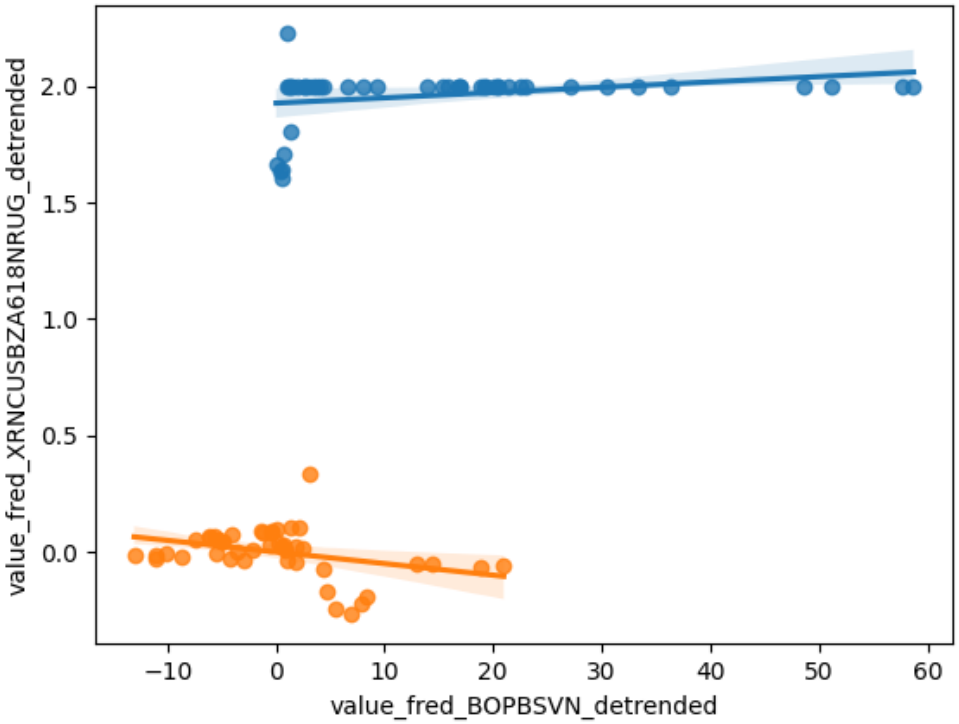



**Figure 2:** Detrended Regression Plot for 2026-10-12